

codex-windows / 019f2174-3b12-7c63-aa29-1dfa56687194

Metadata

```
- Source: `codex-windows`  
- Kind: `codex`  
- Source file: `C:\Users\avidu\.codex\archived_sessions\rollout-2026-07-02T11-41-33-019f2174-3b12-7c63-aa29-1dfa56687194.jsonl`  
- SHA-256: `883daf8df5ba636c1943f281dae9ad7dc048cbd20ee40b58e15c31cbf7a0eab3`  
- Source modified: `2026-07-02T06:11:33+00:00`  
- Imported at: `2026-07-05T16:48:16+00:00`  
- cli_version: `0.142.5`  
- cwd: `C:\Users\avidu\Projects\badminton-highlight-indexer`  
- model_provider: `openai`  
- session_id: `019f2174-3b12-7c63-aa29-1dfa56687194`  
- source: `vscode`
```

Transcript

1. user (2026-07-02T06:11:33.267Z)

Stand up a NEW tracked subproject (owner-requested, exploratory ? for future sessions to build on): package the local badminton-highlight pipeline as a CROSS-PLATFORM DESKTOP APP (Windows/macOS/Linux), with the prerequisite cross-platform port.

VISION: target users are recreational sports enthusiasts (NOT tech-savvy). Their match videos are huge (GBs of 4K). Asking them to upload to a cloud UI is slow, bandwidth-heavy, privacy-fraught, and bloats their drives. The product surface that fits the "local, cheap, efficient" thesis: a DESKTOP APP where the user PLUGS IN the camera's USB/SD card, clicks once, and gets back ONLY the highlight reels ? fully on-device, no upload, no original-video bloat.

YOUR FIRST DELIVERABLE = create the tracked project doc `docs/DESKTOP\_APP\_PLAN.md` following `docs/PROJECT\_DOC\_TEMPLATE.md` (status header · decisions-locked · §7 progress tracker as source-of-truth · definition-of-done), index it one line in `docs/README.md`, and open a PR. Mark everything `[design]` ? nothing is built; this is a scoping doc. Branch from origin/master per the branching-safety convention in CLAUDE.md / ~/.claude/CLAUDE.md (fetch; checkout -b off origin/master; explicit-path staging; ONE PR).

READ FIRST (you have no memory of the originating conversation): `memory/current-state.md` (live state ? the local pipeline is mature: WASB shuttle detector + improved windowing, runs fully-local via `tools/generate\_highlights.py --skip-ai-handoff`), `docs/PLATFORM\_ARCHITECTURE.md` (the PLANNED StorageBackend/ComputeBackend registries ? the right seam for a "LocalDesktop" compute backend), `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` + `docs/COMMERCIALIZATION.md` (strategy/licensing ? commercial-clean MIT/Apache/BSD only; Gemini = inference/optional only, never required, never a training source), `docs/CODE\_MAP.md`.

THE HARD PART ? make this the doc's centerpiece (it's the real "porting to Linux and Mac" subproject): the detector today runs WASB/TrackNet in WSL2 on an NVIDIA GPU (CUDA) ? Windows-specific. ffmpeg is already cross-platform. The port must run the detector NATIVELY per OS (drop the WSL2 dependency): Linux+NVIDIA ? CUDA directly; macOS (Apple Silicon) ? NO CUDA ? Metal/MPS or a CPU path; Windows native (no WSL). THE CRUX IS COMPUTE ON A TYPICAL ENTHUSIAST LAPTOP, which often has NO discrete GPU ? WASB is ~3.4× real-time on a laptop 2070 Super; on CPU it's far slower. So the gating question (the P0 spike): is CPU acceptable for a ~10-min clip? does Apple MPS work for the WASB/TrackNet weights? do we need a smaller/faster owned model (ties OWNED_MODEL_IMPLEMENTATION_PLAN)? a tiered "fast-if-GPU, slow-but-works-if-CPU"? This decides product feasibility ? the doc should call it the make-or-break unknown and propose measuring it.

SCOPE the doc with: vision + why; Phase 1 = the de-WSL native-compute port (per-OS); the app shell options (Tauri (small) vs Electron vs CLI+installer over the existing Python/FastAPI backend); the flow (detect USB/SD insert ? list videos ? on-device proxy + WASB + stitch ? export reels to a user folder, NEVER copy the originals); packaging + code-signing per OS; a §7 tracker (P0 compute-feasibility spike ? P1 de-WSL native detector ? P2 desktop app shell ? P3 packaging/installers); honest limits (feasibility hinges on the compute question; if CPU-only is

too slow on typical laptops, the path may need a small owned model or a hybrid). This is OWNER-GATED implementation ? the scoping doc + (if greenlit) the P0 spike are the deliverable; do NOT build the app without explicit owner approval.

2. assistant (2026-07-02T06:11:33.267Z)

I'll start by following the repo-freshness and branching-safety conventions, then read the required context files. Let me kick these off in parallel.

3. assistant (2026-07-02T06:11:33.267Z)

```
[external_agent_tool_call: Bash]
description: Fetch origin and check git state
command: git fetch origin 2>&1 && echo "---STATUS---" && git status && echo "---CURRENT
BRANCH---" && git branch --show-current && echo "---HEAD---" && git log --oneline -1 && echo "
---ORIGIN/MASTER---" && git log --oneline -1 origin/master
[/external_agent_tool_call]
```

4. assistant (2026-07-02T06:11:33.267Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]
```

5. assistant (2026-07-02T06:11:33.267Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PROJECT_DOC_TEMPLATE.md
[/external_agent_tool_call]
```

6. assistant (2026-07-02T06:11:33.267Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\README.md
[/external_agent_tool_call]
```

7. assistant (2026-07-02T06:11:33.267Z)

```
[external_agent_tool_result]
---STATUS---
On branch feat/long-rally-lens
Your branch is up to date with 'origin/master'.
```

```
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    artifacts/
    docs/KHELSUTRA_PRODUCT_OVERVIEW.md
```

```
nothing added to commit but untracked files present (use "git add" to track)
---CURRENT BRANCH---
feat/long-rally-lens
---HEAD---
a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)
---ORIGIN/MASTER---
a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)
[/external_agent_tool_result]
```

8. assistant (2026-07-02T06:11:33.268Z)

```
[external_agent_tool_result: error]
File content (284.4KB) exceeds maximum allowed size (256KB). Use offset and limit parameters to
read specific portions of the file, or search for specific content instead of reading the whole
file.
```

[/external_agent_tool_result]

9. assistant (2026-07-02T06:11:33.268Z)

[external_agent_tool_result]

```
1      <!--
2      PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
3
4      A "tracked project doc" is how this repo carries substantial design/project work: it can
be
5      ITERATED on, its progress TRACKED, and it is MARKED DONE against explicit deliverables,
then
6      archived. Standardizes the emergent pattern in docs/RALLY_QUALITY_RESEARCH.md §7 and
7      docs/GOLDEN_REGRESSION_FIXTURES.md.
8
9      Rules of thumb:
10     - §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per independently-shippable
PR.
11     - Keep it HONEST: tag each claim [verified] (checked against code), [researched] (needs
sign-off),
12       or [design] (proposed). The doc reflects real shipped state, never aspirational.
13     - It POINTS to detail (code anchors, research artifacts), it does not duplicate it.
14     - Sections marked OPTIONAL are included only when relevant (e.g. Foundation/Threat-model
for
15       research- or security/privacy-heavy work). Delete sections that do not apply.
16     - Move to docs/archives/ once Status = DONE (per the archive policy: only terminal-state
work).
17     -->
18
19     # <Project / Plan Title>
20
21     > **Status:** `DRAFT` ? `<owner-gated? | in progress>` . **Owner:** `<name>` .
**Created:** `<YYYY-MM-DD>` . **Last updated:** `<YYYY-MM-DD>`
22     > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
23     > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
24     > **Relation to existing docs:** `<extends | supersedes | peer-of ?>`
25     > **Honesty note:** mark each claim `[verified]` / `[researched]` / `[design]`. Not
legal/financial advice where applicable.
26
27     ---
28
29     ## 0. TL;DR
30     <2?5 sentences: the goal, the approach, and the single most important caveat. Call out
any finding that is already actionable on its own.>
31
32     ## 1. Problem & goal
33     <Why now; the concrete user-facing outcome; what "good" looks like. Link the code/docs a
reader must read to get current.>
34
35     ## 2. Decisions locked
36     <Table ? capture answered questions so they are not relitigated.>
37
38     | # | Decision | Source / date | Implication |
39     |---|-----|-----|-----|
40     | D1 | ? | ? | ? |
41
42     ## 3. Foundation ? research / prior art *(OPTIONAL)*
43     <Include when the plan rests on external facts: cited research, legal/compliance,
benchmarks, prior-art survey. Mark researched-vs-verified; list any sign-off gates this
creates.>
44
45     ## 4. Design / architecture
46     <The actual plan. Data contracts, diagrams, and a REUSE MAP ? name the existing
```

```

modules/code this builds on so it is clear what is new vs reused.>
47
48    ## 5. Threat model / risk table *(OPTIONAL ? security / privacy / compliance work)*
49    <Who/what is protected vs NOT; residual risks stated plainly.>
50
51    ## 6. Honest limits ? what this does NOT do
52    <The false-confidence guardrail: what a green suite / a DONE status does NOT prove.>
53
54    ## 7. Deliverables & progress tracker  ? **source of truth**
55    Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
56    branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
57
58    | ID | Deliverable | Depends on | Gated? | Status | PR |
59    |----|-----|-----|-----|-----|----|
60    | P0 | ? | ? | No | ? | ? |
61
62    ## 8. Open questions ? owner / external *(split blocking-gates from nice-to-knows; name
63    who decides)*
64    1. ?
65
66    ## 9. Definition of done
67    <Explicit acceptance checklist. When all ticked ? flip `Status ? DONE` and archive.>
68    - [ ] ?
69
70    ## 10. References
71    **Internal code anchors `[verified]`:** `<paths>`. **Internal docs:** `<docs>`.
72    **External:** `<cited sources>`.
73
74    ---
75
76    ### Changelog
77    - `<YYYY-MM-DD>` ? `<what changed>`.
78
79    [/external_agent_tool_result]

```

10. assistant (2026-07-02T06:11:33.268Z)

```

[external_agent_tool_result]
1    - [ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md) ? Latest (v8) vendor
2    labeling guidelines mirrored from the collector's `docs/annotation/`, with the fusion-driven
3    court-calibration / singles-doubles additions marked `? ADDED (fusion)`.
4
5    - [RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) ? **Foundation doc** for
6    closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge
7    diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps,
8    the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-
9    class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an
10   **adversarially-verified, cited external-research synthesis** (TAL/TAS boundary heads, racket-
11   sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap with
12   falsifiable success criteria**, and (§7) a **tracked project plan** ? work-item checklist,
13   sequencing, the single quality **number** attributed to the effort, and a **definition of done**
14   (project completes when all tiers land + the number is recorded). The plan that the
15   [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) log executes against; ties together
16   EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS /
17   OWNED_MODEL_IMPLEMENTATION_PLAN.
18
19   - [RALLY_DETECTION_GAP_CLOSURE.md](RALLY_DETECTION_GAP_CLOSURE.md) ? **Tracked execution
20   doc (2026-06-18)** for closing the remaining rally-detection accuracy gap **per video, before
21   launch**, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it:
22   boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini
23   both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local
24   over-fires), (G3) **rally-END over-extension on the quick-return-after-point** (shuttle still
25   moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-
26   hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally_gate` Gate A
27   wiring, complementary fusion. **Peer** of `RALLY_QUALITY_RESEARCH.md` (execution tracker, not
28   research); §7 is the source of truth.
29
30   - [RALLY_DETECTION_QUALITY_REPORT.md](RALLY_DETECTION_QUALITY_REPORT.md) ? **Technical-

```

paper checkpoint (2026-06-17)** of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.09170.323 at n=11, p=0.002; merge-rate 0.69470.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY\_QUALITY\_RESEARCH.md` + `QUALITY\_ITERATIONS.md`.

```
5      - [EVALUATION.md](EVALUATION.md) ? How we measure and improve quality.
6      - [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md) ? Design (P1 SHIPPED) for A/B + shadow
+ SxS experimentation on rally-detection variants with **zero production regression** ?
experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift,
rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`,
`regression`, `metrics`); implemented in `backend/eval/experiment.py`.
7      - [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) ? **Living log** of rally-detection
quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-
chronological record; terminal-state snapshots get archived under `archives/quality-
iterations/`.
8      - [ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) ? Design (owner-gated,
NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-
scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session
analyzers), a learned **scorer** weights them, and an audi...
[/external_agent_tool_result]
```

11. assistant (2026-07-02T06:11:33.268Z)

Let me read the live state file (it's large, so I'll read the top) and the key strategy docs in parallel.

12. assistant (2026-07-02T06:11:33.268Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]
```

13. assistant (2026-07-02T06:11:33.268Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md
[/external_agent_tool_call]
```

14. assistant (2026-07-02T06:11:33.268Z)

```
[external_agent_tool_result: error]
File content (27122 tokens) exceeds maximum allowed tokens (25000). Use offset and limit
parameters to read specific portions of the file, or search for specific content instead of
reading the whole file.
[/external_agent_tool_result]
```

15. assistant (2026-07-02T06:11:33.268Z)

```
[external_agent_tool_result]
1      # Platform Architecture ? Multi-User "Assistant Coach"
2
3      **Status:** ? Strategy / research ? for review. No code yet.
4      **Date:** 2026-06-07 . **Owner:** avidullu
5
6      > ? **Superseded where it disagrees with
[ `OWNED_MODEL_IMPLEMENTATION_PLAN.md` ](OWNED_MODEL_IMPLEMENTATION_PLAN.md)
7      > (2026-06-09, authoritative). ** Specifically: (1) licensing is **MIT/Apache-2.0/BSD**
(not "Apache-2.0 only");
8      > (2) base model is **Qwen3-VL (Apache)** primary / InternVL3 (MIT) fallback ? **Gemma-4
is AMBER/bench-only**, not a
9      > cleared base; (3) the owned VLM is **not** the per-turn "conversational brain" ?
extraction (TAL head ? later VLM)
```

```

10      > feeds a per-video **event store + text-to-SQL** query layer ("extract once, query
many"); the VLM-answers-each-turn
11      > design is rejected. The §5A model/base/QA-harness detail below predates that and is
kept for context only.
12
13      **Purpose.** Map the path from today's **single-user, local** tool to a **multi-tenant
14      "assistant coach" platform1: users upload videos, get highlights **stored and
15      browsable by history2, and choose **where their data lives** (Google Drive / cloud
16      bucket / local) and **where the compute runs** (our GPU / their VM). The organizing
17      principle is to **decouple the layers** ? identity, ingestion, storage, orchestration,
18      execution/compute, the detection+highlight pipeline, and data-serving ? behind small
19      **pluggable interfaces**, reusing the registry pattern the codebase already uses for
20      segmenters/providers/validators/sports/annotations.
21
22      **Product framing (the north star 3and4 the v1 line).5 This is an **assistant coach**,
23      not just a clipper ? but the strategy is **phased and unambiguous about scope**:
24      - **v1 (concrete, near-term):6 **rally detection ? highlight-reel compilation ? per-
user
25      history ? a coach/user correction loop that feeds the golden set7 ("assistant-coach
26      *lite"). Essentially today's pipeline made multi-user ? this is the real v1
deliverable.
27      - **Mid-term**: **rich rally/shot semantic metadata** (shot type, rally outcome) and the
28      **owned video-QA model** (§5A; §6 P4?P5).
29      - **North-star vision (strictly future ? 8not9 v1):10 **gait / biomechanics, per-athlete
30      identifying templates, and deep coaching feedback under a supervising coach11 (§6 P6).
31      Inspiring, but explicitly deferred ? and the part that raises the **biometric**
privacy
32      bar (§7).
33
34      The buyer is the **academy/coach** (beachhead in
35      [PMF_MARKET_RESEARCH.md])(PMF_MARKET_RESEARCH.md)); **power users self-serve**. Two
36      consequences ripple through the design: (1) the data model keeps an **optional coach ?
37      athlete overlay over a generic core12 (§4.1) ? not baked in; (2) gait/biometrics is
38      **special-category data**, gated and deferred (§7).
39
40      > **Competitive note (validated by research):13 none of Veo / Hudl / SwingVision /
41      > Spiideo offer true BYO-storage or BYO-compute ? they're vendor-cloud SaaS
42      > (SwingVision uniquely runs 14edge15 compute on the user's iPhone). So **"your video,
43      > your storage, your compute16"17 is a genuine differentiator for privacy-sensitive
44      > academies ? and SwingVision proves user-side compute is commercially viable. All four
45      > share the same **teams ? athletes ? clips** data model, which we adopt (§4).
46
47      > Scope: this is a **direction-setting doc**. It does not start implementation; the
48      > first infra slice (an async job model) is the on-hold #16 in
49      > [DEFERRED_HARDENING_PLAN.md](DEFERRED_HARDENING_PLAN.md). Commercial constraints
come
50      > from [COMMERCIALIZATION.md](COMMERCIALIZATION.md) (C4 paid-LLM no-train; C12 SaaS
51      > foundations) and `archives/MONETIZATION_AUDIT.md`.
52
53      ## 1. Current state & gaps (what we're evolving from)
54
55      | Concern | Today | Gap for multi-user |
56      |---|---|---|
57      | Identity / tenancy | none | no users, orgs, auth, or isolation |
58      | Storage | local `filepath` string; output to `output_dir` | no abstraction; ...
[/external_agent_tool_result]

```

16. assistant (2026-07-02T06:11:33.268Z)

Now let me read the live state and locate the key pipeline anchors (WASB/WSL2 detector, stitch, generate_highlights) so the doc is code-grounded.

17. assistant (2026-07-02T06:11:33.268Z)

[external_agent_tool_call: Read]

```
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-  
indexer\memory\current-state.md  
[/external_agent_tool_call]
```

18. assistant (2026-07-02T06:11:33.268Z)

```
[external_agent_tool_result: error]  
File content (27697 tokens) exceeds maximum allowed tokens (25000). Use offset and limit  
parameters to read specific portions of the file, or search for specific content instead of  
reading the whole file.  
[/external_agent_tool_result]
```

19. assistant (2026-07-02T06:11:33.268Z)

```
[external_agent_tool_call: Grep]  
input: {"pattern":"WSL|wsl","path":"C:\\Users\\avidu\\Projects\\badminton-highlight-  
indexer","glob":"*.py","output_mode":"files_with_matches"}  
[/external_agent_tool_call]
```

20. assistant (2026-07-02T06:11:33.268Z)

```
[external_agent_tool_result]  
Found 31 files  
tools\generate_highlights.py  
tests\test_experiment.py  
backend\results.py  
backend\eval\experiment.py  
backend\config\models.py  
tests\test_tracknet_runner.py  
tests\test_fusion_hybrid.py  
backend\pipeline\segmenters\trajectory_hybrid.py  
backend\pipeline\detectors\tracknet_runner.py  
tests\test_tracknet_hybrid.py  
backend\pipeline\detectors\wasb_infer.py  
backend\pipeline\detectors\wasb_runner.py  
tests\test_wasb_cache.py  
tests\test_wsl_tilde_quoting.py  
tests\test_wasb_runner.py  
backend\pipeline\detectors\base.py  
backend\utils\run_telemetry.py  
tests\test_rally_gate.py  
tools\cleanup_caches.py  
tests\test_cleanup_caches.py  
backend\pipeline\segmenters\wasb_hybrid.py  
backend\pipeline\segmenters\tracknet_hybrid.py  
tests\test_reliability_hardening.py  
backend\pipeline\detectors\__init__.py  
tests\test_detector_keying.py  
tests\test_eval_partial_labels.py  
run_all_tests.py  
tests\test_calibrate_local.py  
tests\test_distill_local.py  
tests\test_export_wasb_segments.py  
tests\test_calibrate_wasb.py  
[/external_agent_tool_result]
```

21. assistant (2026-07-02T06:11:33.268Z)

```
[external_agent_tool_call: Read]  
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-  
indexer\memory\current-state.md  
[/external_agent_tool_call]
```

22. assistant (2026-07-02T06:11:33.268Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\wasb_runner.py
[/external_agent_tool_call]
```

23. assistant (2026-07-02T06:11:33.268Z)

```
[external_agent_tool_result]
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-18 (? **R1 MILESTONE LAUNCHED ? #204 MERGED** `67b6927`; flip
13     re-verified at **n=13**, owner-approved). Owner: "verify #204 still stands at n=13, merge if
14     findings stand" ? it stands (stronger), verified 3 ways, merged. **Local-windowing defaults now
15     ON in production: cap=6 + R4** (`max_window_duration 6`, `window_inactivity_end 1.5`,
16     `inactivity_rally_thresh 0.20`, `inactivity_min_density 0.30`). W2 / W1.5 hard-cap / R5 stay
17     default-OFF.
18     - ? **n=13 re-verify (2 new Boxhill clips GX010137/GX010141):** baseline?milestone tolF1
19     **?+0.222, 12/0/1, p=0.0005** (stronger than n=11's 10/0); cap6 beats cap12 **12/0/1,
20     p=0.0005**; cap6?+R4 +0.040, **9/1/3**, p=0.0215 (R4's 1 loss = GX010137, a tolerance artifact ?
21     strict f1 & |?end| both IMPROVE there). Net guard **PASS**, merge_rate 0.651?0.161, **0 per-
22     video regressions**. Both new clips improve (0.448?0.685, 0.333?0.431). tIoU ladder 0.236?0.474.
23     - ? **Corroborated 2 ways:** sibling **#214 `lovo_resweep.py`** matches to 3dp + LOVO
24     cap=6 **13/13** on tIoU; my **5-lens adversarial workflow** `wf_66c34d9d-057` ? **STANDS**
25     (every L00 fold clears; worst-fold drop GX010128 still ?+0.202 p=0.0010 ? no single clip drives
26     it). Reproduce: `scratch/r1_evidence.py`.
27     - ? **#204 shipped clean:** rebased onto master (resolved a config/R5 conflict), 4
28     defaults flipped, 3 default-OFF tests + nightly docstrings updated, mandatory
29     **`docs/R1_MILESTONE_LAUNCH_REPORT.md`** (both rulers + guard + caveats per [[launch-report-
30     convention]]) + QUALITY_ITERATIONS "R1 LAUNCHED". Suite 855 green, cov 80.72%, ruff+mypy clean,
31     7/7 CI ? merged. **#216 re-baselines the nightly** (post-flip `default`==`milestone`; was self-
32     flagging STALE ? re-snapshot n=13, both tolF1 0.359; data-only).
33     - **Caveats (Launch Report):** W1 cap = dominant lever / R4 = marginal +0.040; gate is
34     the NET rate-based guard (strict=False BY DESIGN); GX010141 worst-behaved (future over-merge
35     target); cap=4-vs-6 wobble = noise (LOVO re-fit to cap=4 is *worse*; tIoU firmly cap=6).
36     **Optional follow-up:** add a pinned all-OFF reference config to the nightly to keep the pre-
37     flip floor. ? clean worktrees verify-n13/r1-flip2. [[eval-dual-set-regression]] [[decision-
38     rigor-norm]]
39
40     **Checkpoint:** 2026-06-18 (? **2 MORE BUG-FIX PRs SHIPPED & MERGED ? #212 + #213**;
41     found by adversarial bug-hunt after owner said "if there are any other fixes, send + merge their
42     PRs"). After #209, ran a 2-pass adversarial hunt (by-area ? by-class + completeness critic, ~15
43     agents) over `origin/master`. Pass 1 = 0 findings (clean); pass 2 = 3 confirmed real defects
44     (all owner_gated=false, I verified each against actual code, not trusting the agents):
45     - ? **#212 (`c7baf03`) ? DB migration crash-safety:** v3/v4 used bare non-idempotent
46     `ALTER TABLE ADD COLUMN` while v1/v2/v5 are `IF NOT EXISTS`. A DDL ALTER auto-commits
47     independently of the `with conn:` scope, so an interrupted upgrade (after ALTER, before the end-
48     of-block `PRAGMA user_version` bump) leaves the column present + version stale ? next open re-
49     runs the ALTER ? `OperationalError: duplicate column name` ? **`__init__` throws forever =
50     permanently bricked DB**. Fix: `_add_column_idempotent()` swallows ONLY 'duplicate column name'.
51     Regression test proven to FAIL without the fix. Fully testable in CI (pure sqlite).
52     - ? **#213 (`bd33085`) ? scripts unwrap segmenter Result:** `scripts/run_phase3_eval.py`
53     (`--segmenter` default==`motion`, the documented baseline) + `run_process_and_stitch.py` did
54     `len()/truthiness on a raw `Success`/`Failure` dataclass (no `__len__`) ? `TypeError` on t...
55 [/external_agent_tool_result]
```

24. assistant (2026-07-02T06:11:33.268Z)

```
[external_agent_tool_result]
1     """WASB shuttle-trajectory runner (WSL2 subprocess adapter).
2
3     Windows-side counterpart to backend/detectors/wasb_infer.py. It:
4         1. stages the video into the WSL filesystem (fast local reads, not /mnt),
5         2. copies wasb_infer.py into the WASB repo's src/ dir,
6         3. runs it in the WSL `wasb` conda env -> a Frame,Visibility,X,Y trajectory CSV,
7         4. reuses the (model-agnostic) trajectory->action-window logic from tracknet_runner.
8
9     So the only WASB-specific bits live here; the windowing + CSV parsing are shared
10    with the TrackNet path. Verified inference core: wasb_infer.py (see its docstring).
11    """
12    import os
13    import shlex
14    import logging
15    import subprocess
16    from dataclasses import dataclass
17    from typing import Dict, Any, Optional, Tuple
18
19    # Reuse the model-agnostic helpers ? trajectory shape + windowing are identical.
20    # parse_trajectory_csv / trajectory_to_action_windows are @staticmethods on
21    TrackNetRunner.
22    from backend.pipeline.detectors.tracknet_runner import to_wsl_mnt_path, TrackNetRunner
23    from backend.pipeline.detectors.base import DetectorRunner, WslCommandMixin,
24    wsl_tilde_quote
25
26    logger = logging.getLogger(__name__)
27
28    # Windows path to the inference wrapper we stage into WSL.
29    _INFER_WIN = os.path.join(os.path.dirname(os.path.abspath(__file__)), "wasb_infer.py")
30
31    @dataclass
32    class WasbConfig:
33        """Resolved settings for a WASB WSL run (built from config.json ->
34        indexing.wasb)."""
35        distro: str = "Ubuntu"
36        repo_dir: str = "~/models/WASB-SBDT"
37        conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
38        conda_env: str = "wasb"
39        weights_path: str = "~/models/WASB-
40        SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
41        sport: str = "badminton"
42        wsl_stage_dir: str = "~/clips"
43        timeout_sec: int = 1800 # 30 min; a hung GPU/conda call is killed (0 = no timeout)
44        keep_frames: bool = False # retain staged video + frame cache after success (debug
45        only)
46        min_free_gb: float = 0.0 # warn if WSL free space below this before a run (0 =
47        off)
48        # FAST PATH (default OFF until verified): decode the staged video on the fly inside
49        # WSL and feed frames straight to the detector, skipping the slow per-frame PNG
50        # extraction that dominates a long clip. Output is bit-identical to the PNG path
51        # (PNG is lossless), but this stays opt-in until the owner's side-by-side confirms
52        it.
53        stream_video: bool = False
54
55    @classmethod
56    def from_indexing_cfg(cls, idx_cfg: Dict[str, Any]) -> "WasbConfig":
57        w = (idx_cfg or {}).get("wasb", {}) or {}
58        return cls(
59            distro=w.get("wsl_distro", "Ubuntu"),
60            repo_dir=w.get("repo_dir", "~/models/WASB-SBDT"),
61            conda_sh=w.get("conda_sh", "~/miniconda3/etc/profile.d/conda.sh"),
```

```

56         conda_env=w.get("conda_env", "wasb"),
57         weights_path=w.get("weights_path",
58                             "~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar"),
59         sport=w.get("sport", "badminton"),
60         wsl_stage_dir=w.get("wsl_stage_dir", "~/clips"),
61         timeout_sec=int(w.get("timeout_sec", 1800)),
62         keep_frames=bool(w.get("keep_frames", False)),
63         min_free_gb=float(w.get("min_free_gb", 0.0)),
64         stream_video=bool(w.get("stream_video", False)),
65     )
66
67
68     class WasbRunner(WslCommandMixin, DetectorRunner):
69         def __init__(self, cfg: WasbConfig):
70             self.cfg = cfg
71
72         def healthcheck(self) -> Tuple[bool, str]:
73             """Verify the WSL `wasb` env imports torch and sees a GPU. Returns (ok, msg)."""
74             cmd = (f"conda activate {self.cfg.conda_env} && "
75                   f"python -c \"import torch; print('torch', torch.__version__, "
76                   f"'cuda', torch.cuda.is_available())\"")
77             try:
78                 res = self._wsl_bash(cmd)
79             except:
80                 pass
81             return (True, "")
82
83     [/external_agent_tool_result]

```

25. assistant (2026-07-02T06:11:33.268Z)

Let me ground the de-WSL crux: read the detector base (WSL mixin), the actual inference core, the local entrypoint, and locate the packaging item + owned-model tie-in.

26. assistant (2026-07-02T06:11:33.269Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py
[/external_agent_tool_call]

```

27. assistant (2026-07-02T06:11:33.269Z)

```

[external_agent_tool_result]
1     """Abstract base for detector runners (TrackNet, WASB, future native, etc.).
2
3     This completes the Detector Runner Abstraction (low-regret item 6 continuation / item 1
4     of the post-observability re-prioritization) to de-risk the WSL/conda dependency for
5     the high-accuracy shuttle trajectory substrate used by tracknet_hybrid / wasb_hybrid.
6
7     The ABC lets us:
8     - Keep the current WSL adapters working unchanged for Windows users.
9     - Swap in (or A/B test) a Linux-native implementation later with zero changes to
10    the hybrid segmenters or the model-agnostic windowing brain.
11    - Provide a single place to evolve the healthcheck / error contract and thread
12    detector readiness into per-video reports.
13
14    Contract (deliberately matches what the concrete runners + hybrids + tests already did):
15    - run_predict(video_win_path: str, output_win_dir: str) -> Optional[str]
16      Run detector, return the Windows-side path to the produced trajectory CSV
17      (e.g. ".../clip_predict.csv" or ".../clip_wasb.csv"), or None on any failure.
18      The caller (hybrid) is responsible for output dir; impls may stage the video
19      into WSL fs for perf.
20
21    - healthcheck() -> tuple[bool, str]
22      Return (ok, message). Never raises. "ok" decides whether to proceed to run_predict.
23      Message is logged (and can be surfaced to reports or UX in future).

```

```

24         Intended to be cheap (env existence, import, GPU visibility, weights present).
25
26     Existing behavior is 100% preserved. The only additions are the inheritance and
27     the explicit common type for future implementers.
28
29     How to add a new runner (for docs / future contributors):
30     1. Subclass DetectorRunner in a new or existing module under pipeline/detectors/.
31     2. Implement run_predict and healthcheck (return (bool, str) tuple).
32     3. If it produces the same CSV schema (Frame,Visibility,X,Y), reuse or re-export
33         TrackNetRunner.parse_trajectory_csv and .trajectory_to_action_windows (they are
34         deliberately static and model-agnostic). If the new detector needs different
35         windowing, it can implement its own or we generalize later.
36     4. Lazy-import heavy native deps inside the methods (so importing the segmenter
37         registry never pulls torch/tf on a machine that doesn't have them).
38     5. Update the two hybrid segmenters (or introduce a detector= choice) to accept
39         DetectorRunner instances (they already do the right thing via duck-typing + the
40         type hints we add here).
41     6. Add unit tests that exercise the contract (isinstance, healthcheck shape, run
42         returns str|None) using mocks for the external bits ? see test_tracknet_runner.py.
43     7. No change to eval/calibration code paths (they consume the CSV + the static
44         windowing helpers directly).
45
46     Optional Linux-native path (the main de-risk goal):
47     A LinuxNativeRunner (or ONNXRunner) would:
48     - Skip all wsl / _stage / conda activate.
49     - Load weights locally (torch or onnx) or call a native binary.
50     - Write the exact same CSV format so trajectory_to_action_windows works verbatim.
51     - healthcheck can just check "weights file exists and torch importable + CUDA?".
52     This removes the WSL requirement for Linux users / future SaaS workers while
53     keeping the exact same candidate quality characteristics (or better, once native
54     weights are available).
55
56     Error surfacing into reports:
57     The healthcheck message is already logged by the hybrids on failure. The reporting
58     harvest sees the outcome as "0 candidates" + detector name. When we wire richer
59     segmenter Results (or pass detector_health into emit_indexer_report context), the
60     (ok, msg) can be recorded explicitly under segmentation.details or a new detector
61     stage. The ABC makes that future change local to one place.
62
63     See also:
64     - backend/pipeline/segmenters/{tracknet_hybrid,wasb_hybrid}.py (the two callers)
65     - tests/test_tracknet_runner.py (contract + pure logic tests; hybrids mock the runner)
66     - docs/T...
[/external_agent_tool_result]

```

28. assistant (2026-07-02T06:11:33.269Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\detectors\wasb_infer.py
[/external_agent_tool_call]

```

29. assistant (2026-07-02T06:11:33.269Z)

```

[external_agent_tool_result]
1     """WASB shuttle-trajectory inference on an arbitrary frame sequence (runs in WSL).
2
3     WASB-SBDT only ships *dataset* evaluation; this is a thin wrapper that reuses its
4     own building blocks (ImageDataset transform + detector + online tracker) to run on
5     any folder of extracted frames and emit a trajectory CSV in the exact shape the
6     indexer's `tracknet_runner.parse_trajectory_csv` already consumes:
7
8         Frame,Visibility,X,Y
9
10    It must run INSIDE the WSL `wasb` conda env, from the WASB-SBDT `src/` dir (it

```

```

11     imports WASB's `detectors`/`trackers`/`dataloaders`). The Windows-side adapter
12     (`wasb_runner.py`) stages this file + the frames into WSL and invokes it.
13
14     Resumable across reboots
15     -----
16     The GPU detector pass (one forward per sliding window ? ~21k for a 6-min clip) is
17     the slow, expensive part; the CPU tracker pass that turns detections into a
18     trajectory is cheap and *stateful* (it must see every frame in order, so it can't
19     resume mid-stream). We exploit that split:
20
21     * The detector pass writes its per-window raw outputs incrementally to
22     ``<cache>/det_raw.jsonl`` (flushed+fsync'd every ``--flush-every`` windows) and
23     records progress in ``<cache>/manifest.json`` (atomic write). A reboot loses at
24     most ``--flush-every`` windows of GPU work instead of the whole pass.
25     * On restart we replay the cached windows, skip the DataLoader ahead to the first
26     un-cached window, and continue. Once every window is cached, the detector pass
27     is skipped entirely and we just re-run the (cheap, deterministic) tracker over
28     the full ordered cache -> trajectory CSV. So a lost trajectory CSV costs seconds,
29     not the GPU hour.
30
31     The cache is keyed by the output path (``<out>_wasbcache/`` by default), lives next
32     to the output (NOT in %TEMP%), and is invalidated automatically if the frames dir,
33     weights, sport, window size, or frame count change.
34
35     Usage (in WSL, from ~/models/WASB-SBDT/src):
36     python wasb_infer.py --frames_dir /path/frames --weights
37     ../pretrained_weights/wasb_badminton_best.pth.tar \
38     --out /path/traj.csv [--sport badminton] [--limit N] [--cache-dir DIR] [--flush-
39     every 1000] [--fresh]
40     """
41     import os
42     import os.path as osp
43     import csv
44     import glob
45     import json
46     import argparse
47     import logging
48     from collections import defaultdict
49     from contextlib import contextmanager
50
51     logger = logging.getLogger(__name__)
52
53     CACHE_VERSION = 1
54     _DET_FILE = "det_raw.jsonl"
55     _MANIFEST_FILE = "manifest.json"
56
57     def _build_cfg(weights: str, sport: str):
58         """Compose the WASB eval config via Hydra, overriding model/weights/device."""
59         from hydra import compose, initialize
60         with initialize(config_path="configs", version_base=None):
61             cfg = compose(config_name="eval", overrides=[
62                 f"dataset={sport}",
63                 "model=wasb",
64                 f"detector.model_path={weights}",
65                 "runner.device=cuda",
66                 "runner.gpus=[0]", # this box has a single GPU; default config assumes
67             ])
68         return cfg
69
70     def _frame_id(path: str) -> int:
71         [/external_agent_tool_result]

```

30. assistant (2026-07-02T06:11:33.269Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tools\generate_highlights.py
[/external_agent_tool_call]
```

31. assistant (2026-07-02T06:11:33.269Z)

```
[external_agent_tool_result]
1      """Batch highlight-reel runner ? orchestrates "generate highlights for these videos
2      into this folder" requests with CLEAN, NON-DESTRUCTIVE, resumable execution.
3
4      Why this exists (lessons paid for the hard way):
5      - **Never delete other work.** Reels + per-video run.logs are written to a LOCAL work
dir
6      first, then COPIED into the destination (atomically: temp name -> os.replace). The
7      destination folder is only ever ADDED to ? we never bulk-delete it. (A bulk `rm` of a
8      Google-Drive-synced folder once lost finished reels + logs.)
9      - **Drive-sync safe.** Generating directly inside a Google Drive folder races the sync
10     client (deleted files resurrect, partial writes sync). Generate locally; publish once.
11     - **Idempotent re-runs.** Each video's prior DB segments are cleared before processing
12     (add_video is an UPSERT and does NOT clear children), so a re-run never accumulates
13     duplicate rallies. Belt-and-suspenders with VideoCompiler's overlap-merge (PR #180).
14     - **Resumable & concurrency-friendly.** A video whose reel already exists in the
15     destination is skipped (unless --force); launching a second request never undoes the
16     first's published reels. The Gemini path also isolates its chunk workdir per video
17     (PR for output/chunks/<video_id>) so concurrent videos don't collide.
18
19     Usage:
20     python -m tools.generate_highlights --videos-dir "G:\\...\\June 12 2026" \
21     --out "G:\\...\\June 12 2026\\Highlights" --segmenter gemini [--force]
22     python -m tools.generate_highlights --video A.MP4 --video B.MP4 --out <dir> ...
23     """
24     from __future__ import annotations
25
26     import argparse
27     import logging
28     import os
29     import shutil
30     import sys
31     import traceback
32     from typing import Any, Dict, List, Optional, cast
33
34     from dotenv import load_dotenv
35
36     sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
37
38     from backend.config import load_config
39     from backend.infrastructure.database import Database
40     from backend.pipeline.segmenters.base import segmenter_registry
41     from backend.pipeline.stitcher.compiler import VideoCompiler
42     from backend.results import Failure
43     from backend.utils.hashing import compute_video_id
44
45     logger = logging.getLogger("highlights_runner")
46
47     # Detectors that run in WSL can't read Google Drive / UNC via /mnt -> localize first.
48     _WSL_DETECTORS = {"wasb_hybrid", "tracknet_hybrid", "yolo_hybrid"}
49     _VIDEO_EXTS = (".mp4", ".MP4", ".mov", ".MOV", ".mkv", ".avi")
50
51
52     def _atomic_publish(local_path: str, dest_path: str) -> None:
53         """Copy local_path -> dest_path so the destination never sees a partial file.
54
55         Copy to a temp sibling, then os.replace (atomic on the same volume). Never deletes
```

```

56         anything else in the destination folder.
57         """
58         os.makedirs(os.path.dirname(dest_path), exist_ok=True)
59         tmp = dest_path + ".publishing.tmp"
60         shutil.copy2(local_path, tmp)
61         os.replace(tmp, dest_path)
62
63
64     def generate_highlights_batch(video_paths: List[str], out_dir: str, *, segmenter: str,
65                                   local_work_dir: str, db_path: str, force: bool = False,
66                                   skip_ai_handoff: bool = False, wasb_stream: bool = False,
67                                   detect_from_paths: Optional[List[str]] = None) -> dict:
68         """Generate one highlight reel per input video into out_dir. Returns a summary dict.
69
70         Non-destructive: only ADDS reels/logs to out_dir (atomic publish); never deletes.
71
72         skip_ai_handoff: run the trajectory-hybrid segmenters fully-local (no Gemini
handoff);
73         candidate windows become the rallies. wasb_stream: have the WASB detector decode
the
74         video on the fly instead of dumping ~46k PNGs (output-preserving fast path,
#178).
75         detect_from_paths: optional list parallel to video_paths. When given, the detector
runs on
76         detect_from_paths[i] (a...
[/external_agent_tool_result]

```

32. assistant (2026-07-02T06:11:33.269Z)

Let me check for any prior desktop/packaging mentions and the owned-model size tie-in, so the doc cross-links accurately and avoids duplication.

33. assistant (2026-07-02T06:11:33.269Z)

```

[external_agent_tool_call: Grep]
input: {"pattern":"desktop|Tauri|Electron|packaging|installer|code.?sign|PyInstaller|app
shell","path":"C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs","output_mode":"files_with_matches","-i":true}
[/external_agent_tool_call]

```

34. assistant (2026-07-02T06:11:33.269Z)

```

[external_agent_tool_result]
Found 13 files
docs\\NEXT_STEPS.md
docs\\CODE_MAP.md
docs\\archives\\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
docs\\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
docs\\PLATFORM_ARCHITECTURE.md
docs\\BACKLOG.md
docs\\COMMERCIALIZATION.md
docs\\DEFERRED_HARDENING_PLAN.md
docs\\VIDEO_LOCALITY_MODEL.md
docs\\archives\\past_projects\\low-regret-moves\\LOW_REGRET_MOVES_PLAN.md
docs\\archives\\checkpoints\\2026-06-strategy-foundation\\README.md
docs\\archives\\checkpoints\\2026-06-scaffolding-complete\\README.md
docs\\archives\\COMMERCIAL_READINESS_REVIEW.md
[/external_agent_tool_result]

```

35. assistant (2026-07-02T06:11:33.269Z)

```

[external_agent_tool_call: Grep]
input: {"pattern":"#7|packaging|distribut","path":"C:\\Users\\avidu\\Projects\\badminton-
highlight-indexer\\docs\\DEFERRED_HARDENING_PLAN.md","output_mode":"content","-i":true,"-C":2}

```

```
[/external_agent_tool_call]
```

36. assistant (2026-07-02T06:11:33.269Z)

```
[external_agent_tool_result]
1:# Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs, Packaging (COMPLETED)
2-
3-***STATUS: COMPLETED** (2026-06-14). This planning doc's three items are all dispatched:
4-
5-- **16 Async job model ? DONE** (PR #87; hardened by the self-scheduling drain loop in #149).
6:- **7 Packaging modernization ? DONE** (PR #167: PEP 621 `pyproject.toml`, `setup.py` ?
`scripts/doctor.py`, `indexer` console entry point, CI switched to `pip install -e .[dev]`).
7-- **13 Train/eval separation guardrails ? moved to [BACKLOG.md](BACKLOG.md)** (?; the
concrete `splits.py` / `assert_disjoint` match-level-split design is preserved there).
8-
[/external_agent_tool_result]
```

37. assistant (2026-07-02T06:11:33.269Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"MPS|Metal|Apple
Silicon|ONNX|CPU|onnxruntime|CoreML|quantiz","path":"C:\\Users\\avidu\\Projects\\badminton-
highlight-indexer\\docs\\OWNED_MODEL_IMPLEMENTATION_PLAN.md","output_mode":"content","-
i":true,"head_limit":40}
[/external_agent_tool_call]
```

38. assistant (2026-07-02T06:11:33.269Z)

```
[external_agent_tool_result]
36:weights); a *base* is the metal you forge. You pick one framework and feed it any clean base.
91: Python ? **export across the boundary** (ONNX/Core ML/ExecuTorch/GGUF) ? a thin device
shell (Swift/Kotlin/C++/Rust)
94:- Escalation order if a real Python CPU hot path ever appears: free-threading ?
numpy/vectorize ? Cython/Py03 ? only then a
[/external_agent_tool_result]
```

39. assistant (2026-07-02T06:11:33.269Z)

`docs/VIDEO\_LOCALITY\_MODEL.md` sounds directly on-thesis. Let me read it and the owned-model export snippet.

40. assistant (2026-07-02T06:11:33.269Z)

```
[external_agent_tool_call: Read]
file: C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\VIDEO_LOCALITY_MODEL.md
[/external_agent_tool_call]
```

41. assistant (2026-07-02T06:11:33.269Z)

```
[external_agent_tool_result]
1      # Video locality ? where the bytes live (mental model + open questions)
2
3      **Status:** THINKING DOC (owner-requested, 2026-06-12) ? a shared mental model for
4      the upload/storage problem, with the open questions named. **No implementation is
5      greenlit by this doc** (per CLAUDE.md, both tracks stay owner-gated).
6      **Companions:** `UI_PROPOSAL.md` (B2/B3 upload/download), `HOSTING_PLAN.md`,
7      `PLATFORM_ARCHITECTURE.md` §3 (future StorageBackend/ComputeBackend registries),
8      collector ADR-002 (the frame-identical proxy contract this doc leans on).
9
10     ## 0. The questions that triggered this (and the short answers)
11
12     ***"Do we store the video they upload?"** Yes ? today the chunked-upload flow
13     assembles the full file under `security.upload_dir` (default `output/uploads`) on
14     our box, registers it by MD5, and nothing ever deletes it (retention policy is an
```

```

15     acknowledged gap ? BACKLOG). Processing also strews derived artifacts (chunks,
16     frames, trajectories, WSL temp) and, on the Gemini path, uploads ?1280px
17     re-encoded chunks to Google's Files API.
18
19     """How would they upload a 10 GB file?""" Today they can't: `/api/upload/init`
20     rejects anything over `security.max_upload_mb` (default **4096 MB**), and even
21     under the cap the file moves as sequential ?90 MB parts through a free-tier edge
22     whose body limit forced that part size. For scale: 10 GB ? 30?45 min of 4K phone
23     video (or ~2?3 h at 1080p); on a typical Indian uplink that's **~1 h at 20 Mbps,
24     ~4.5 h at 5 Mbps** ? before retries. Browser-tab uploads of that size are
25     miserable everywhere; expecting academies to manage S3/GCS buckets is a
26     non-starter. **The 10 GB problem is real and the current design doesn't solve it.**
27
28     """Can we make it so there is NO upload?""" Mostly ? because of $1.
29
30     ## 1. The central insight: timestamps are the product; the video is dead weight
31
32     Walk the pipeline and ask what each stage actually consumes:
33
34     - The **Gemini segmenter** never sees the original ? it re-encodes chunks down to
35       **1280 px** (`indexing.ai_chunk_scale_width`) before uploading them to Google.
36     - The **detectors/featurizer** consume downsampled frames and trajectories.
37     - The only consumer of the original full-res bytes is the **highlight compiler**,
38       and only to cut clips ? a pure-ffmpeg operation that can run **wherever the
39       original lives**.
40     - What the customer receives is `{rally intervals, report, reel}`. Intervals +
41       report are **kilobytes**. Given the intervals, the reel can be cut on the
42       machine that already has the original.
43
44     So: **any design that moves the 10 GB is moving the wrong thing.** The question
45     is never "how do we upload the video" but "what is the smallest artifact that
46     must cross the network for this tier of service."
47
48     ## 2. The locality ladder
49
50     From most to least bytes leaving the academy's Windows desktop. Sizes are rough
51     ratios for a 4K source ? measure on a real file before committing to any tier.
52
53     | | What crosses the network | ~Bytes (10 GB source) | What it needs | Status |
54     |---|---|---|---|---|
55     | **L3 ? full upload** (today's flow) | everything | 10 GB (blocked: 4 GB cap) | paid
56     edge tier + resumable uploads + server disk + retention policy | built, capped, wrong for big
57     files |
58     | **L2 ? proxy upload** | a timeline-identical ?1080p re-encode | ~1?2.5 GB | a tiny
59     client prep step (ffmpeg) ? **the exact frame-identical contract we already built and verified
60     as collector `prep-annotation-proxy`** (same fps, same frame count, parity-checked, MD5?MD5
61     manifest) | contract proven; client wrapper + server acceptance = new work |
62     | **L1 ? chunks/candidates upload** | only the ?1280px AI chunks (or just candidate
63     windows) the Gemini path needs anyway | ~0.3?1 GB | client-side chunk+downscale (ffmpeg, CPU);
64     server becomes a **GPU-less orchestrator** | the server half exists (it's the gemini segmenter's
65     own flow, relocated) |
66     5...
67     [/external_agent_tool_result]

```

42. assistant (2026-07-02T06:11:33.269Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-
indexer\docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
[/external_agent_tool_call]

```

43. assistant (2026-07-02T06:11:33.269Z)

```

[external_agent_tool_result]
78     Apache/BSD shuttle/ball detectors for tennis/TT/pickleball/padel are not yet

```

```

identified.** Sequence by public-data
79     richness: **badminton ? tennis/TT ? pickleball/padel.**
80     - ? **Single-model-multisport-temporal-generalization is an OPEN HYPOTHESIS**, not
proven ? our corpus is badminton-only,
81     and the evidence once cited for it (RacketVision) was *refuted on verification* (it's
ball-tracking, not temporal). The
82     concept is sound on first principles (rally signal = a racquet-sport invariant) but
must be validated once a 2nd-sport
83     corpus exists. Don't cite RacketVision as evidence.
84
85     ## Language ? Python everywhere; on-device runtime is a deferred, layer-local export
shell
86     - **Training/eval = Python-locked, and that's fine** (PyTorch/ms-swift/ActionFormer);
offline/batch, no latency hot path.
87     - **Platform = no Python hot path to escape** ? the heavy bytes/GPU already run out-of-
process (ffmpeg, WASB-in-WSL, cloud
88     GPU per-job); the control plane is I/O-bound. **Reject a Go/Rust hot-path split** ?
single-digit-ms wins dwarfed by
89     ~40-min GPU jobs, and it would fracture the one codebase holding the QA-state + eval
harness + provider registry.
90     - **On-device (deferred) = the only place a 2nd language appears, and it doesn't dictate
our dev language:** train in
91     Python ? **export across the boundary** (ONNX/Core ML/ExecuTorch/GGUF) ? a thin device
shell (Swift/Kotlin/C++/Rust)
92     runs the artifact with zero Python. **The one concrete constraint now: train with
export-clean ops** so the compression
93     target stays reachable without re-architecting. (On-device base stays on the
Qwen3-VL-4B Apache lineage ? *not* Gemma-3-270M.)
94     - Escalation order if a real Python CPU hot path ever appears: free-threading ?
numpy/vectorize ? Cython/PyO3 ? only then a
95     separate service. **None exists today.**
96
97     ---
98
99     ## Phase 0 ? Foundation (start now, per-milestone greenlight; $0, local, no cloud/DPA)
100
101     ### M0.1 ? The labeled-corpus spine **+ the event-index contract** (the moat) .
*greenlight item*
102     Canonical JSONL row per rally (corpus) **and** the queryable per-video event-index
schema (the conversational seam)
103     ? same row, wired to a store. Fields: `{video_id(MD5), rally_ordinal, sport, start, end,
ending_reason/fault_tags,
104     shot_count, label_type, source(human|pseudo|vendor), consent/training_opt_in, split(BY
MATCH), trajectory_ref,
105     labeled_window, derived_clip_ref, confidence}`. 3 transcoders (heuristic/LogReg . TAL
ActivityNet-JSON . VLM clip?QA),
106     one scorer spine (`metrics.match_intervals`). Authored in `sports-data-collector`
(system-of-record; extends PR #13).
107     **Reserve fault/event taxonomy now.** Acceptance: round-trip test; split-by-match
enforced; **no Gemini-sourced rows in any training view**.
[/external_agent_tool_result]

```

44. assistant (2026-07-02T06:11:33.270Z)

I have everything I need. `VIDEO\_LOCALITY\_MODEL.md` (its L0 "fully local" tier + OQ5 packaging) and the owned-model plan's "export across the boundary" section are the perfect anchors, and `backend/pipeline/detectors/base.py` literally pre-specifies the de-WSL native runner. Let me set up the branch off `origin/master`.

45. assistant (2026-07-02T06:11:33.270Z)

```

[external_agent_tool_call: Bash]
description: Create branch off origin/master
command: git checkout -b docs/desktop-app-plan origin/master 2>&1 && echo "---VERIFY---" && git
branch --show-current && git log --oneline -1

```

[/external_agent_tool_call]

46. assistant (2026-07-02T06:11:33.270Z)

[external_agent_tool_result]
branch 'docs/desktop-app-plan' set up to track 'origin/master'.
Switched to a new branch 'docs/desktop-app-plan'
---VERIFY---
docs/desktop-app-plan
a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)
[/external_agent_tool_result]

47. assistant (2026-07-02T06:11:33.270Z)

[external_agent_tool_result]
Chapter marked: "Desktop app plan doc". Continue your current work.
[/external_agent_tool_result]

48. assistant (2026-07-02T06:11:33.270Z)

Now I'll write the tracked project doc.

49. assistant (2026-07-02T06:11:33.270Z)

[external_agent_tool_call: mcp_ccd_session__mark_chapter]
input: {"title":"Desktop app plan doc","summary":"Author docs/DESKTOP_APP_PLAN.md scoping doc,
index it, open PR"}
[/external_agent_tool_call]

50. assistant (2026-07-02T06:11:33.270Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DESKTOP_APP_PLAN.md
[/external_agent_tool_call]

51. assistant (2026-07-02T06:11:33.270Z)

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DESKTOP_APP_PLAN.md (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

52. assistant (2026-07-02T06:11:33.270Z)

The doc is written. Now let me index it in `docs/README.md`. The top block is the rally-quality track (no header); a desktop/product scoping doc fits best under a new section. Let me add it before "Practical / ongoing".

53. assistant (2026-07-02T06:11:33.270Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\README.md
[/external_agent_tool_call]

54. assistant (2026-07-02T06:11:33.270Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

55. assistant (2026-07-02T06:11:33.270Z)

Both files are ready. Let me re-verify the branch (per branching-safety) and commit with explicit paths only.

56. assistant (2026-07-02T06:11:33.270Z)

```
[external_agent_tool_call: Bash]
description: Re-verify branch state before staging
command: echo "---BRANCH---" && git branch --show-current && echo "---HEAD (should be
mine/origin master tip)---" && git log --oneline -1 && echo "---ONLY MY COMMITS vs origin/master
(should be empty pre-commit)---" && git log --oneline origin/master..HEAD && echo "---STATUS---"
&& git status --short
[/external_agent_tool_call]
```

57. assistant (2026-07-02T06:11:33.270Z)

```
[external_agent_tool_result]
---BRANCH---
docs/desktop-app-plan
---HEAD (should be mine/origin master tip)---
a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)
---ONLY MY COMMITS vs origin/master (should be empty pre-commit)---
---STATUS---
  M backend/eval/rally_seg_eval.py
  M docs/README.md
?? artifacts/
?? docs/DESKTOP_APP_PLAN.md
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
[/external_agent_tool_result]
```

58. assistant (2026-07-02T06:11:33.270Z)

I notice `backend/eval/rally\_seg\_eval.py` is modified ? I did **not** touch it; it's sibling/owner WIP that appeared in the tree (exactly the shared-checkout scenario the convention warns about). I'll stage **only** my two explicit paths and leave everything else untouched.

59. assistant (2026-07-02T06:11:33.270Z)

```
[external_agent_tool_call: Bash]
description: Stage explicit paths and verify staged diff
command: git add docs/DESKTOP_APP_PLAN.md docs/README.md && echo "---STAGED (should be ONLY my 2
files)---" && git diff --cached --name-only && echo "---README staged diff (confirm only my
added section)---" && git diff --cached docs/README.md
[/external_agent_tool_call]
```

60. assistant (2026-07-02T06:11:33.271Z)

```
[external_agent_tool_result]
warning: in the working copy of 'docs/DESKTOP_APP_PLAN.md', LF will be replaced by CRLF the next
time Git touches it
---STAGED (should be ONLY my 2 files)---
docs/DESKTOP_APP_PLAN.md
docs/README.md
---README staged diff (confirm only my added section)---
diff --git a/docs/README.md b/docs/README.md
index c2e7266..b9b9ca2 100644
--- a/docs/README.md
+++ b/docs/README.md
@@ -10,6 +10,9 @@
```

- GOLDEN_REGRESSION_FIXTURES.md ? **Design** (owner-gated, NOT started): **video-free**, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only

public tier). Carries a cited ****IN/US/EU legal memo**** (derived-data IP/privacy), a threat model, and a ****\$7 progress tracker + definition-of-done****. Extends
GOLDEN_DATA_SHARING.md.

+++ Product / platform plans

+ - DESKTOP_APP_PLAN.md ? ****Tracked scoping doc (2026-06-18, owner-gated, [design] ? NOTHING built)**** package the local pipeline as a ****cross-platform desktop app**** for non-technical players ? plug in the camera's USB/SD card, click once, get back ****only the highlight reels****, fully on-device (no upload, no original-video bloat). The centerpiece is the ****de-WSL native-compute port**** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its ****make-or-break unknown****: is CPU acceptable on a typical ****no-GPU**** enthusiast laptop (WASB is ~3.4x real-time on a laptop 2070S)? \$7 tracker = ****P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers****. Realizes the ****L0 fully-local**** tier of VIDEO_LOCALITY_MODEL.md; is the ****LocalDesktop ComputeBackend**** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

+
Practical / ongoing
- TRACKNET_WSL_SETUP.md
- Golden set docs ([GOLDEN_SET_`](
[/external_agent_tool_result]

61. assistant (2026-07-02T06:11:33.271Z)

Staged content is exactly my two files. Committing and pushing.

62. assistant (2026-07-02T06:11:33.271Z)

[external_agent_tool_call: Bash]
description: Commit scoping doc and push branch
command: git commit -m '@'
docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port

New tracked project doc DESKTOP_APP_PLAN.md (owner-gated, all [design] ? nothing built). Vision: package the mature local pipeline (WASB + improved windowing + ffmpeg stitch, already runnable via `generate\_highlights.py --skip-ai-handoff`) as a desktop app for non-technical recreational players ? plug in the camera's USB/SD card, click once, get back ONLY the highlight reels, fully on-device (no upload, no original-video bloat).

Centerpiece = the de-WSL native-compute port (run WASB natively per-OS: Linux/Windows CUDA, macOS MPS/CPU, no WSL2), grounded in the DetectorRunner ABC that already anticipates a Linux-native runner. The make-or-break unknown is compute on a typical NO-GPU enthusiast laptop (WASB ~3.4x real-time on a laptop 2070S); the doc names a P0 feasibility spike to measure CPU/MPS before any app is built.

Follows PROJECT_DOC_TEMPLATE (status header, decisions-locked, \$7 tracker as source-of-truth: P0 spike -> P1 native detector -> P2 app shell -> P3 packaging, definition-of-done). Realizes the L0 tier of VIDEO_LOCALITY_MODEL and the LocalDesktop ComputeBackend of PLATFORM_ARCHITECTURE §3.2; ties the smaller export-clean owned model as one answer to the compute crux. Indexed in docs/README.md. Owner-gated implementation ? only this scoping doc is in flight.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>
 '@ 2>&1 && echo "---PUSH---" && git push -u origin docs/desktop-app-plan 2>&1
[/external_agent_tool_call]

63. assistant (2026-07-02T06:11:33.271Z)

[external_agent_tool_result: error]
Exit code 2

```
[docs/desktop-app-plan f721274] @ docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port
2 files changed, 167 insertions(+)
create mode 100644 docs/DESKTOP_APP_PLAN.md
/usr/bin/bash: eval: line 33: syntax error near unexpected token `('
/usr/bin/bash: eval: line 33: `USB/SD card, click once, get back ONLY the highlight reels, fully on-device (no'
[/external_agent_tool_result]
```

64. assistant (2026-07-02T06:11:33.271Z)

The commit landed but with a mangled message ? I used PowerShell here-string syntax (`@'...'@`) inside the **Bash** tool (POSIX sh), so the `` became literal and the parens broke parsing. Let me check and fix the commit message.

65. assistant (2026-07-02T06:11:33.271Z)

```
[external_agent_tool_call: Bash]
description: Inspect the committed message
command: git log --oneline -1 && echo "---FULL MSG---" && git log -1 --format=%B
[/external_agent_tool_call]
```

66. assistant (2026-07-02T06:11:33.271Z)

```
[external_agent_tool_result]
f721274 @ docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port
---FULL MSG---
@
docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port
```

New tracked project doc DESKTOP_APP_PLAN.md (owner-gated, all [design] ? nothing built). Vision: package the mature local pipeline (WASB + improved windowing + ffmpeg stitch, already runnable via `generate\_highlights.py --skip-ai-handoff`) as a desktop app for non-technical recreational players ? plug in the cameras
[/external_agent_tool_result]

67. assistant (2026-07-02T06:11:33.271Z)

I'll write the commit message to a temp file and amend with `-F` (clean, no shell-quoting pitfalls).

68. assistant (2026-07-02T06:11:33.271Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\_desktop_commit_msg.txt
[/external_agent_tool_call]
```

69. assistant (2026-07-02T06:11:33.271Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\_desktop_commit_msg.txt (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

70. assistant (2026-07-02T06:11:33.271Z)

```
[external_agent_tool_call: Bash]
description: Amend commit message from file and verify
command: git commit --amend -F scratch/_desktop_commit_msg.txt 2>&1 && rm -f scratch/_desktop_commit_msg.txt && echo "---AMENDED SUBJECT---" && git log --oneline -1 && echo "---VERIFY ONLY MY COMMIT vs origin/master---" && git log --oneline origin/master..HEAD && echo "---STILL ONLY 1 FILE-SET CHANGED---" && git show --stat --oneline HEAD | head -8
[/external_agent_tool_call]
```

71. assistant (2026-07-02T06:11:33.271Z)

```
[external_agent_tool_result]
[docs/desktop-app-plan b9ff235] docs(desktop): scope a cross-platform desktop app + the de-WSL
native-compute port
Date: Thu Jun 18 13:18:53 2026 +0530
2 files changed, 167 insertions(+)
create mode 100644 docs/DESKTOP_APP_PLAN.md
---AMENDED SUBJECT---
b9ff235 docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port
---VERIFY ONLY MY COMMIT vs origin/master---
b9ff235 docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port
---STILL ONLY 1 FILE-SET CHANGED---
b9ff235 docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port
docs/DESKTOP_APP_PLAN.md | 164 ++++++
docs/README.md | 3 +
2 files changed, 167 insertions(+)
[/external_agent_tool_result]
```

72. assistant (2026-07-02T06:11:33.271Z)

Clean ? only my commit, only my two files. Now push and open the PR.

73. assistant (2026-07-02T06:11:33.271Z)

```
[external_agent_tool_call: Bash]
description: Push branch to origin
command: git push -u origin docs/desktop-app-plan 2>&1
[/external_agent_tool_call]
```

74. assistant (2026-07-02T06:11:33.271Z)

```
[external_agent_tool_result]
remote:
remote: Create a pull request for 'docs/desktop-app-plan' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/docs/desktop-app-
plan
remote:
branch 'docs/desktop-app-plan' set up to track 'origin/docs/desktop-app-plan'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      docs/desktop-app-plan -> docs/desktop-app-plan
[/external_agent_tool_result]
```

75. assistant (2026-07-02T06:11:33.271Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\badminton-highlight-
indexer\scratch\_desktop_pr_body.md (file state is current in your context ? no need to Read it
back)
[/external_agent_tool_result]
```

76. assistant (2026-07-02T06:11:33.271Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\_desktop_pr_body.md
[/external_agent_tool_call]
```

77. assistant (2026-07-02T06:11:33.271Z)

```
[external_agent_tool_call: Bash]
description: Open PR against master
command: gh pr create --base master --head docs/desktop-app-plan --title "docs(desktop): scope a
cross-platform desktop app + the de-WSL native-compute port" --body-file
scratch/_desktop_pr_body.md 2>&1 && rm -f scratch/_desktop_pr_body.md
```

[/external_agent_tool_call]

78. assistant (2026-07-02T06:11:33.271Z)

[external_agent_tool_result]
<https://github.com/avidullu/badminton-highlight-indexer/pull/219>
[/external_agent_tool_result]

79. assistant (2026-07-02T06:11:33.271Z)

PR is open: [#219](<https://github.com/avidullu/badminton-highlight-indexer/pull/219>). Let me record a micro-checkpoint in the live memory (per the checkpoints working agreement) ? a new tracked subproject + PR is a meaningful checkpoint.

80. assistant (2026-07-02T06:11:33.271Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]

81. assistant (2026-07-02T06:11:33.271Z)

[external_agent_tool_result: error]
<tool_use_error>File has been modified since read, either by the user or by a linter. Read it again before attempting to write it.</tool_use_error>
[/external_agent_tool_result]

82. assistant (2026-07-02T06:11:33.271Z)

The memory file changed under me (concurrent session). Let me re-read the top and re-apply.

83. assistant (2026-07-02T06:11:33.272Z)

[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]

84. assistant (2026-07-02T06:11:33.272Z)

[external_agent_tool_result]
1 ---
2 name: current-state
3 description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not only on request). What just landed, what's next, open decisions."
4 metadata:
5 node_type: memory
6 type: project
7 originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8 ---
9
10 # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12 **Checkpoint:** 2026-06-18 (? **R1 MILESTONE LAUNCHED ? #204 MERGED** `67b6927`; flip re-verified at **n=13**, owner-approved). Owner: "verify #204 still stands at n=13, merge if findings stand" ? it stands (stronger), verified 3 ways, merged. **Local-windowing defaults now ON in production: cap=6 + R4** (`max\_window\_duration 6`, `window\_inactivity\_end 1.5`, `inactivity\_rally\_thresh 0.20`, `inactivity\_min\_density 0.30`). W2 / W1.5 hard-cap / R5 stay default-OFF.
13 - ? **n=13 re-verify (2 new Boxhill clips GX010137/GX010141):** baseline?milestone tolF1 **?+0.222, 12/0/1, p=0.0005** (stronger than n=11's 10/0); cap6 beats cap12 **12/0/1, p=0.0005**; cap6?+R4 +0.040, **9/1/3**, p=0.0215 (R4's 1 loss = GX010137, a tolerance artifact ? strict f1 & |?end| both IMPROVE there). Net guard **PASS**, merge_rate 0.651?0.161, **0 per-

video regressions**. Both new clips improve (0.448±0.685, 0.333±0.431). tIoU ladder 0.236±0.474.
14 - ? **Corroborated 2 ways:** sibling **#214 `lovo\_resweep.py`** matches to 3dp + LOVO
cap=6 **13/13** on tIoU; my **5-lens adversarial workflow** `wf\_66c34d9d-057` ? **STANDS**
(every L00 fold clears; worst-fold drop GX010128 still ?+0.202 p=0.0010 ? no single clip drives
it). Reproduce: `scratch/r1\_evidence.py`.
[/external_agent_tool_result]

85. assistant (2026-07-02T06:11:33.272Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]

86. assistant (2026-07-02T06:11:33.272Z)

[external_agent_tool_result]
The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

87. assistant (2026-07-02T06:11:33.272Z)

Let me do a final verification ? PR state and that the working tree is clean apart from the pre-existing sibling WIP.

88. assistant (2026-07-02T06:11:33.272Z)

[external_agent_tool_call: Bash]
description: Verify PR and clean working tree
command: echo "---PR---" && gh pr view 219 --json
number,title,state,baseRefName,headRefName,files --jq
'{number,title,state,base:.baseRefName,head:.headRefName,files:[.files[]].path}}' 2>&1 && echo "
---WORKING TREE (sibling WIP + untracked only; scratch msgs gone)---" && git status --short
[/external_agent_tool_call]

89. assistant (2026-07-02T06:11:33.272Z)

[external_agent_tool_result]
---PR---
{
 "base": "master",
 "files": [
 "docs/DESKTOP_APP_PLAN.md",
 "docs/README.md"
],
 "head": "docs/desktop-app-plan",
 "number": 219,
 "state": "OPEN",
 "title": "docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port"
}
---WORKING TREE (sibling WIP + untracked only; scratch msgs gone)---
M backend/eval/rally_seg_eval.py
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
[/external_agent_tool_result]

90. assistant (2026-07-02T06:11:33.272Z)

Done. The tracked subproject is scoped and the PR is open.

What landed ? [PR #219](https://github.com/avidullu/badminton-highlight-indexer/pull/219) (docs-only, owner

****New doc:**** docs/DESKTOP_APP_PLAN.md, indexed one-line in docs/README.md under a new "Product / platform plans" section. Follows `PROJECT\_DOC\_TEMPLATE` (status header · D1?D6 decisions-locked · \$7 tracker as source-of-truth · definition-of-done).

****The vision it scopes:**** package the mature local pipeline (WASB + improved windowing + ffmpeg stitch, already runnable via `generate\_highlights.py --skip-ai-handoff`) as a cross-platform

desktop app for non-technical players ? *plug in the card, click once, get back only the reels*, fully on-device, no upload, no original-video bloat. It's framed as realizing the **L0 fully-local** tier of [VIDEO_LOCALITY_MODEL.md](docs/VIDEO_LOCALITY_MODEL.md) and the **`LocalDesktop` ComputeBackend** of [PLATFORM_ARCHITECTURE.md](docs/PLATFORM_ARCHITECTURE.md) §3.2.

The centerpiece (the real "port to Linux/Mac" subproject): the de-WSL native-compute port. I grounded this in code ? backend/pipeline/detectors/base.py already documents a `LinuxNativeRunner`/`ONNXRunner` as "the main de-risk goal," the WSL plumbing is isolated in a separate mixin, and the torch inference core in [wasb_infer.py](backend/pipeline/detectors/wasb_infer.py) just needs lifting out of WSL (identical CSV contract ? windowing/stitch untouched).

The make-or-break unknown, made the doc's spine: compute on a typical *no-GPU* enthusiast laptop. WASB is ~3.4× real-time on a laptop 2070S (measured); CPU is far slower, Apple has no CUDA. **P0** is a feasibility spike that measures CPU/MPS for a ~10-min clip and returns go/no-go before any app code ? and explicitly names the fork: if CPU-only is too slow, the path needs a smaller exported owned model (ties [OWNED_MODEL_IMPLEMENTATION_PLAN.md](docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md)) or a tiered fast/slow design.

§7 tracker: P0 compute spike ? P1 de-WSL native detector ? P2 app shell (Tauri vs Electron vs CLI+installer) ? P3 packaging/code-signing per OS. Only P0 is un-gated; P1?P3 are owner-greenlit *after* P0. Licensing locked to MIT/Apache/BSD + an LGPL ffmpeg build (no GPL x264).

Process notes

- Branched off `origin/master` (`a010667`) per the branching-safety convention; staged **explicit paths only**.
- ?? A sibling/owner WIP modification to `backend/eval/rally\_seg\_eval.py` appeared in the shared checkout mid-task ? I left it **untouched** (not staged, not committed).
- (Minor: my first commit's message got mangled because I used PowerShell here-string syntax inside the Bash tool; amended cleanly via `-F`.)
- Recorded a micro-checkpoint in `memory/current-state.md`.

Owner gate stands: this is the scoping doc only. The P0 spike ? and anything past it ? needs your explicit greenlight before I build.